

# Session 4

## Generative AI: retrieval and evaluation

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What you build today: Project 3, the policy-grounded response service.

# The job

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- Answer policy questions from the current corpus, not from memory
- Cite the passage used, and mean it
- Refuse when the corpus does not cover the question
- Never show an internal document to a customer
- Prove all of that with numbers that survive a model swap

## Structured output is the contract

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- Everything downstream consumes a validated object, never prose
- JSON mode fails on long outputs and nested optionals
- Tool calling as a schema fails when the model declines the tool
- Schema in the prompt, validation in code, one repair: works on every vendor

# Chunking is a decision

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| Strategy         | Keeps                          | Loses                     |
|------------------|--------------------------------|---------------------------|
| Fixed windows    | nothing to configure           | headings, metadata        |
| Heading-aware    | section, title, date, audience | little on structured docs |
| Sentence windows | local context                  | structure                 |

Policies have headings and effective dates. Chunk on the headings, carry the metadata.

## Tokenise sensitive fields first

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- Email, account id, card, phone, key: replaced by a stable token
- Before the index, before the prompt, before the log
- Stable within a document so the model can still refer to "[ACCOUNT\_1]"
- A test proves the values never reach a chunk

## Hybrid retrieval

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- Dense finds paraphrases, blurs "99.95" and "sla-2023"
- BM25 finds exact terms, misses paraphrases
- Reciprocal rank fusion merges the lists, no weight to tune
- Filter by metadata after retrieval: current only, customer audience

# Reranking

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- A cross-encoder reads query and chunk together
- Reorders thirty candidates, not the corpus
- Most of the quality in this session, and most of the latency
- Measure both before you keep it

## Citations are enforced in code

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1. Tell the model which ids it may cite
2. Check every citation against the context ids
3. Drop invented ones, count them
4. No valid citation left: refuse

Deterministic. Citation validity must be exactly 1.0 or the code is wrong.

## Refusal is an outcome

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| Path                       | Reason                                      |
|----------------------------|---------------------------------------------|
| Retrieval too weak         | <code>weak_retrieval</code> , no model call |
| Model says not covered     | <code>model_unanswerable</code>             |
| No valid citation survived | <code>no_valid_citation</code>              |

Measured against cases that must refuse, including internal-only answers.

## Evaluate retrieval and generation separately

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| Metric              | Computed by                |
|---------------------|----------------------------|
| Recall@k, MRR       | deterministic              |
| Faithfulness        | the Judge, with a rubric   |
| Citation validity   | deterministic, must be 1.0 |
| Refusal correctness | deterministic              |
| Latency, cost       | the Session 1 meter        |

A bad answer from good retrieval is a different bug from a bad answer from bad retrieval.

## The judge is a model too

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- A different, stronger model than the workhorse, fixed version
- Judges drift, prefer their own style, and can be argued with
- Spot-check a sample by hand every cohort
- Never treat a judge score as ground truth

## The regression gate

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- Write the baseline once
- Every run compares; a drop beyond tolerance fails CI
- On an engagement, the harness is what survives the vendor change

## Change one variable

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Around the room, one each: chunk size 200 or 1000, k 4 or 12, hybrid off, reranker off, min score up.

Rerun the harness. Put your row on the board. Most changes you were sure would help will not.

## Acceptance criteria

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1. Sensitive fields never reach a chunk; dates and percentages do
2. Superseded and internal chunks filtered by metadata
3. Citation validity exactly 1.0 on the golden set
4. Recall@8 at or above 0.85 and correct refusal on at least 10 of 12 unanswerable cases
5. A report with measured cost and p95 latency, and a passing regression gate

`make session04` reports every test passed.

## Questions worth asking

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- Chunk size from 350 to 1000: which metric moves first, and which way?
- The judge says 0.5 on an answer you think is right. Who is right?
- The answer is only in the internal escalation matrix. What does the service say, and what does the log say?